Exercise 1.
Out of 1000 adults 550 are women and 450 men. The standard deviation of the weight of men is 10kg and we know that half of the men weighs less than 80kg and as for women, these data are 15kg and 60kg. Suppose that the weights of men and women are normally distributed.

a) What is the mean value of the weight of a random person?
b) What is the standard deviation of the weight of a random person?
c)What is the probability that a random peson weighs less than 70kg?!

Exercise 2.
Suppose X~N(mu,sigma). Find the density function of Y=exp(X), where exp is the exponential function with base e (=2.71828+).

Exercise 3. Suppose that your computer can generate only random numbers that are uniformly distributed on (0,1).
How would you transform them to obtain random numbers that are standard normally distributed?

Exercise 4. Let Z and X be independent random variables such that Z~N(0,1), X~Bern(0.5).
Show that
a) XZ ~N(0,1);
b) Z+XZ is NOT normally distributed.

(This is interesting, since it is known that the sum of two independent normally distributed random variables is again normally distributed)

Exercise 5. Consider two components and three types of shocks. A type 1 shock causes component 1 to fail, a type 2 shock causes component 2 to fail, and a type 3 shock causes both components 1 and 2 to fail. The times until shocks 1, 2, and 3 occur are independent exponential random variables with respective rates λ1, λ2, and λ3. Let Xi denote the time at which component i fails, i = 1, 2. The random variables X1,X2 are said to have a joint bivariate exponential distribution. Find P(X1 >s and X2 > t).

6. You want to sell your house at the first offer above $200,000.
The offers come independently and they are normally distributed with mean $180,000 and standard deviation $20,000. On the average how many offers will you deny?